THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The converse of the proposition "If two angles are vertically opposite, then they are equal." is: 1

(a) If two angles are not equal, then they are not vertically opposite.
(b) If two angles are equal, then they are vertically opposite.
(c) Vertically opposite angles are equal.
(d) If two angles are not vertically opposite, then they are not equal.

2. The converse of the proposition "If $x = 3$, then $x^2 = 9$." is: 1

(a) If $x = -3$, then $x^2 = 9$. **(b)** If $x^2 \neq 9$, then $x \neq 3$.
(c) If $x \neq 3$, then $x^2 \neq 9$. **(d)** If $x^2 = 9$, then $x = 3$.

3. Which of the following is a counterexample to the proposition "If a number is a multiple of 4, then it is a multiple of 8."? 1

(a) 16 **(b)** 8 **(c)** 20 **(d)** 24

4. Proposition: "If $x > 5$, then $x > 2$." Its converse is false because: 1

(a) $x = 3$ satisfies $x > 2$ but not $x > 5$ **(b)** no number is greater than 2
(c) $x = 10$ is greater than 5 **(d)** $x = 6$ satisfies both

5. A triangle has sides 9 cm, 12 cm and 16 cm. By the converse of the Baudhāyana–Pythagoras theorem it is: 1

(a) right-angled, since $a^2 + b^2 = c^2$
(b) not right-angled, since $a^2 + b^2 \neq c^2$
(c) right-angled, since the sides are in increasing order
(d) right-angled, since $a + b > c$

6. The number 100 has: 1

(a) 11 factors; it is a perfect square **(b)** 11 factors; it is not a perfect square
(c) 10 factors; it is not a perfect square **(d)** 9 factors; it is a perfect square

7. The number of factors of 35 is: 1

(a) 3 **(b)** 2 **(c)** 4 **(d)** 5

8. The sentence "Y when X" conveys the same meaning as: 1

(a) X and Y are false **(b)** If Y then X
(c) If not X then not Y **(d)** If X then Y

9. Which pair of statements are converses of each other and both true? 1

(a) "If it rains, the road is wet" and "If the road is wet, it has rained"
(b) "If a triangle is equilateral, then all its angles are equal" and "If all angles of a triangle are equal, then it is equilateral"
(c) "If a number ends in 0, then it is even" and "If a number is even, then it ends in 0"
(d) "If $x = 2$, then $x^2 = 4$" and "If $x^2 = 4$, then $x = 2$"

10. **Assertion (A):** The converse of the Baudhāyana–Pythagoras theorem is true. 1
Reason (R): If $a^2 + b^2 = c^2$, a right triangle with legs a, b has hypotenuse c, so the given triangle is congruent to it by SSS.
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Give a counterexample to "If n is prime, then $2^n - 1$ is prime." 2

12. Show that the number 9 is a counterexample to the proposition "If a number is odd, then it is prime". 2

13. Write the converse of "If a triangle is right-angled with sides a, b, c (hypotenuse c), then $a^2 + b^2 = c^2$." Is it true? 2

14. Is the proposition "If a quadrilateral is a rectangle, then its diagonals are equal" true? Write its converse and give a counterexample for it. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why a number that has exactly one repeated factor pair must have an odd number of factors. 3

16. Write the converse of "If a and b are perfect squares, then ab is a perfect square". Is the proposition true? Is the converse true? Justify. 3

17. Find a counterexample to each: 3
(i) If n is prime, then $n + 4$ is prime.
(ii) If n is a multiple of 5, then n ends in 5.

OR

State the converse of "If n is divisible by 60, then it is divisible by both 5 and 12". Is the converse true? Justify. 3

18. Show that "If n is divisible by 4 and by 6, then it is divisible by 24" is false. Give a counterexample and explain why the idea '4×6 = 24' fails. 3

OR

A class says "If it is Monday, our school has an assembly." Write the converse and explain why it may not be true. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write the converse of each proposition and decide whether each of the two statements is true. 5
(i) If $x = y$, then $x^3 = y^3$.
(ii) If $x = y$, then $x^2 = y^2$.

OR

Show that the triangle with vertices on the grid $A(0, 0)$, $B(7, 0)$, $C(0, 24)$ is right-angled using the Baudhāyana–Pythagoras converse, and find its hypotenuse and area. 5

20. **Case study:** A carpenter wants a corner to be exactly 90°. He marks 3 units on one wall edge, 4 units on the other and checks that the distance across is 5 units. 5
(i) Which result is used?
(ii) Show the calculation.
(iii) Is the converse needed or the theorem itself?
(iv) Will 5-12-13 work too? Verify.

OR

(i) Prove that "If n is the square of a prime p, then n has exactly 3 factors." 5
(ii) State and prove the converse.

END OF QUESTION PAPER — Total: 40 Marks